Homework 3

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April 17, 2026

2.4.2 Model Satisfaction

Sentence: $\phi \triangleq \forall x \exists y \exists z (P(x,y) \land P(z,y) \land (P(x,z) \to P(z,x)))$

Evaluation of Natural Number Models

- **(a)** $\mathcal{M} : (\mathbb{N}, <)$: Let x be arbitrary. Choose $y = x+1$ and $z = x$. $P(x,y)$ is true as $x < x+1$. $P(z,y)$ is true as $x < x+1$. For the implication, $P(x,z)$ is $x < x$ (False), making $F \to F$ true. Thus, $\mathcal{M} \models \phi$.
- **(b)** $\mathcal{M}' : (\mathbb{N}, P(m,n) \textbf{ iff } n = 2m)$: For any x, choose $y = 2x$ and $z = x$. This satisfies $P(x,y)$ and $P(z,y)$. Since $x = z$, the final conjunct holds vacuously. Thus, $\mathcal{M}' \models \phi$.
- **(c)** $\mathcal{M}'' : (\mathbb{N}, m < n+1)$: Choosing $y = x+1$ and $z = x$ satisfies $P(x,y)$ $(x < x+2)$ and $P(z,y)$ $(x < x+2)$. The implication holds as $P(x,x)$ is $x < x+1$, but $P(z,x)$ is also $x < x+1$ (True $\to$ True). Thus, $\mathcal{M}'' \models \phi$.

2.4.3 Identity and Predicates

Goal: Find a model satisfying $\forall x \neg P(x,x)$ and one that does not.

- **Model** $\mathcal{M} \models \forall x \neg P(x,x)$: Let $D = \{1,2\}$ and $P^{\mathcal{M}} = \{(1,2),(2,1)\}$. No element relates to itself.
- **Model** $\mathcal{M}' \not\models \forall x \neg P(x,x)$: Let $D = \{1\}$ and $P^{\mathcal{M}'} = \{(1,1)\}$. Here $P(1,1)$ is true, so the negation is false.

2.4.7 Semantic Entailment Proof

Show: $\forall x \neg \phi \models \neg \exists x \phi$

Proof. Assume $\mathcal{M} \models \forall x \neg \phi$. By definition, for every $a \in D^{\mathcal{M}}$, $\mathcal{M} \models \neg \phi[a/x]$, meaning ϕ is false for all a. Suppose $\mathcal{M} \models \exists x \phi$. This requires at least one a where $\phi[a/x]$ is true. This is a direct contradiction. Thus, the model must satisfy $\neg \exists x \phi$. $\square$

2.4.8 Semantic Entailment

Show: $\forall x P(x) \vee \forall x Q(x) \models \forall x(P(x) \vee Q(x))$

Proof. Assume $\mathcal{M} \models \forall x P(x) \vee \forall x Q(x)$. If $\mathcal{M} \models \forall x P(x)$, then for any a, $P(a)$ is true, making $P(a) \vee Q(a)$ true. If $\mathcal{M} \models \forall x Q(x)$, then for any a, $Q(a)$ is true, making $P(a) \vee Q(a)$ true. In both cases, $\forall x(P(x) \vee Q(x))$ holds. □

2.4.10 Counter-justification

Is $\forall x(P(x) \vee Q(x)) \models \forall x P(x) \vee \forall x Q(x)$**? No.**

Let $D = \{a, b\}$ with $P^{\mathcal{M}} = \{a\}$ and $Q^{\mathcal{M}} = \{b\}$. Every element in D satisfies either P or Q, so the left side is true. However, $\forall x P(x)$ is false (fails for b) and $\forall x Q(x)$ is false (fails for a). Thus, $T \models F$ is false.

2.4.12 Validity and Counter-examples

(a) Symmetry vs. Reflexivity

Invalid. Let $D = \{1\}, S = \{(1, 1)\}$. Premise $S(1, 1) \to S(1, 1)$ is True, but conclusion $\neg S(1, 1)$ is False.

(c) Universal-Existential Shift I

Valid. Assume $\forall x(P(x) \to \exists y Q(y))$. For an arbitrary x_0, if $P(x_0)$ is false, the implication $P(x_0) \to Q(y)$ is vacuously true for any y. If $P(x_0)$ is true, there exists y such that $Q(y)$ is true, making the implication true.

(d) Universal-Existential Shift II

Valid. Assume $\forall x \exists y(P(x) \to Q(y))$. For any x, there is a y where $P(x) \to Q(y)$ holds. If $P(x)$ is true, $Q(y)$ must be true for that y. Thus $\exists y Q(y)$ is satisfied.

(f) Antisymmetry vs. Irreflexivity

Invalid. Let $D = \{0\}, S = \{(0, 0)\}$. Premise $(S(0, 0) \to 0 = 0)$ is true, but conclusion $\neg S(0, 0)$ is false.

(h) Global Equivalence

Invalid. Let $D = \{0, 1\}, P = \{0\}$. For $x = 0, y = 1$, the conjunct $(P(0) \to P(1))$ is $T \to F$, which is false.

(i) Bi-implication Distribution

Valid. Assume $\forall x(P(x) \leftrightarrow Q(x))$ and $\forall x P(x)$. For any a, $P(a)$ is true. Since $P(a) \leftrightarrow Q(a)$, then $Q(a)$ must be true. Therefore, $\forall x Q(x)$ holds.